Math 534 H - Spring 2026 - Lecture 18.

Q: In what sense does

$$u(x,t) = \frac{1}{\sqrt{4\pi k t}} \int_{-\infty}^{\infty} e^{-(x-y)^2/4kt} \cdot \phi(y)\, dy$$

satisfies that $u(x,0) = \phi(x)$? In the sense of Distributions

We cannot plug in $t=0$! We must take $\lim_{t \to 0^+} u(x,t) = \phi(x)$.

$\hookrightarrow$ prove

We recall from Lecture 17 that

$$\lim_{t \to 0^+} \underbrace{S(x,t)}_{\text{Heat kernel.}} = \delta(0)$$ Dirac delta at the origin.

What this means if ϕ that is smooth

then $\delta : \phi \longmapsto \phi(0)$

More generally δ_{x_0} = Dirac delta at x_0

This means

$$\delta_{x_0} : \phi \longrightarrow \phi(x_0)$$

Then when you take $\lim_{t \to 0} u(x,t)$ you get

this limit equals $\int \underbrace{S(x-y,t)}_{\xrightarrow{t \to 0^+}} \phi(y)\,dy$.

$$\text{"} \int \delta(x-y)\,\phi(y)\,dy \qquad (*)$$

(But this is not an integral any more but is a <u>pairing</u> (AS DISTRIBUTIONS) and simply means that you are recenter the Dirac delta at x and $(*)$

$$\delta_x : \phi \longrightarrow \phi(x) \,;$$ in other

words $\boxed{\int \delta(x-y)\,\phi(y) = \phi(x).}$

Let's do a nice application of the Uniqueness of solutions to the heat equation on $\mathbb{R}$.

In Problem 9 Section 2.4 you are asked to find the solution to

$$\begin{cases} u_t - k\,u_{xx} = 0 \\ u(x,0) = x^2 \end{cases}$$

$(x \in \mathbb{R},\ k>0,\ t>0)$

They tell you to solve it not by using the

heat kernel representation but by proceeding as follows:

1) Note u_{xxx} also solves the heat equation but with IC equal to 0.

2) Integrating 3 times; from 1) you get that the general form for

$$u(x,t) = \underline{A(t)x^2 + B(t)x + C(t)}$$

3) Take formula in 2) you plug in

$$u_t - k\,u_{xx} = 0 \longrightarrow A(t), B(t), C(t)$$

you get $\begin{cases} A(t) = A_0 & \text{constant in } t \\ B(t) = B_0 & \text{constant in } t. \\ C(t) = 2kA_0 t + C_0 & C_0 \text{ constant} \end{cases}$

$A'(t) = 0$

$B'(t) = 0$

$C'(t) = 2kA(t)$

4) Finally you use the initial condition $u(x,0) = x^2$ to show $A_0 = 1$, $B_0 = 0$, $C_0 = 0$.

5) All in all you get

$$\boxed{u(x,t) = x^2 + 2kt} \quad (\dagger\dagger)$$

What happens if we use the heat kernel representation? We set

$$u(x,t) = \frac{1}{\sqrt{4\pi k t}} \int_{-\infty}^{\infty} e^{-\frac{(x-y)^2}{4kt}} \cdot \overbrace{y^2}^{\phi(y)} \, dy \qquad (+)$$

Next let's make some change of variables

Call $p = \dfrac{x-y}{\sqrt{4kt}} \Rightarrow y = x - \sqrt{4kt}\, p.$

$$dp = -\frac{dy}{\sqrt{4kt}}.$$

Back to (+) and substitute:

$$u(x,t) = -\frac{1}{\sqrt{\pi}} \int_{\infty}^{-\infty} \underbrace{\left(x - \sqrt{4kt}\, p\right)^2}_{y^2} e^{-p^2}\, dp$$

Next develop the square and switch the limits of integration to get

$$u(x,t) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} \left(x^2 - 2\sqrt{4kt}\, xp + 4ktp^2\right) e^{-p^2}\, dp$$

This gives 3 integrals:

$$u(x,t) = \frac{1}{\sqrt{\pi}} \left[x^2 \underbrace{\int_{-\infty}^{\infty} e^{-p^2}\, dp}_{\sqrt{\pi}} - 2\sqrt{4kt}\, x \underbrace{\int_{-\infty}^{\infty} p e^{-p^2}\, dx}_{0} \right.$$

$$+ \; 4kt \underbrace{\int_{-\infty}^{\infty} p^2 e^{-p^2} \, dp}_{= I} \Big]$$

We are a bit stuck. How can we compute this? One way is to use Uniqueness of Heat eq. and (††) in page 6.

All in all we get that

$$u(x,t) = x^2 + \frac{4kt}{\sqrt{\pi}} I$$

But we already know from before that

$u(x,t) = x^2 + 2kt$; and solutions are UNIQUE

Hence $$\cancel{x^2} + \frac{4kt}{\sqrt{\pi}} \cdot I = \cancel{x^2} + 2kt$$

$$\Rightarrow \boxed{I = \frac{\sqrt{\pi}}{2}}$$

$$\Rightarrow \boxed{\int_{-\infty}^{\infty} p^2 e^{-p^2}\, dp = \frac{\sqrt{\pi}}{2}.}$$

Remark: Yes, as Aarnav said in class we could have integrated by parts and use the calculation for $\int_{-\infty}^{\infty} e^{-p^2} dp$ that gave $\sqrt{\pi}$. But

Once you have obtained (tt) page 6 the point is that uniqueness of the Heat equation gives you the value of I immediately.

We continue now with Chapter 4.

Recall you need to **read** Handouts 9 and 10 (Course webpage) which cover Sections 3.2 and 3.3 in Chapter 3. of Strauss' book.

Please Do! and ask me any questions you have.

<u>Ch. 4</u> Let's start with the BIVP for the wave equation with homogeneous Dirichlet boundary conditions on $[0, L]$

$$\begin{cases} \text{EQ.} & u_{tt} - c^2 u_{xx} = 0 \quad x \in [0, L],\ t > 0 \\ \text{BC.} & u(0,t) = 0 = u(L,t) \ \forall t > 0 \\ \text{IC.} & u(x,0) = \phi(x);\ u_t(x,0) = \psi(x) \end{cases}$$

① First we consider $\begin{cases} u_{tt} - c^2 u_{xx} = 0 \\ u(0,t) = 0 = u(L,t) \end{cases}$

Idea is to make an <u>ansatz</u> (guess) and look for solution of the form

$$u(x,t) = X(x)\,T(t) \qquad \begin{pmatrix} T(t) \neq 0 \\ X(x) \neq 0 \end{pmatrix}$$

(by separation of variables)

Plug into the EQ. we get

$$X(x)\,\underbrace{T''(t)}_{\text{diff. in } t} - c^2\left(\underbrace{X''(x)}_{\text{diff in } x}\cdot T(t)\right) = 0$$

$$\Longleftrightarrow \quad \underbrace{\frac{T''(t)}{c^2\,T(t)}}_{\text{dep. only on } t} = \underbrace{\frac{X''(x)}{X(x)}}_{\text{dep. only on } x} \quad \rightarrow \text{ Constant!}$$

MUST BE CONSTANT in t and x